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A Hybrid Multimodal Deep Learning Framework for Alzheimer's Disease Classification and Survival Analysis Using Image and Clinical Data

Sadiya Naz Kaladgi^{a[0009-0000-6610-2702]}, Deepa Kalmath^a, Shivaraj Yadavannavar^a, Umesh Annigeri^a, Salma Shahapur^b, Amey Muchandi^b, Rajashri Khanai^b

^{a,b}Department of Computer Science Engineering, KLE Technological University, Belagavi, Karnataka, 590008, India

Abstract

To overcome the limitations of single-modality data in the diagnosis of the Alzheimer's disease (AD), the present study presents a Hybrid Multimodal Deep Learning Framework that unites structural MRI scans and patient-specific clinical variables. We recover rich imaging and clinical embeddings using EfficientNet-B0 and BrainFusionNet and combine them to classify stages of AD and the survival prognosis of the disease using Cox Proportional Hazards and DeepSurv models. We also compared DenseNet-201 and an ensemble model. Tests on the OASIS dataset indicate that EfficientNet-B0 can clearly be used to improve unimodal approaches: 97.5% with EfficientNet-B0, 97.0% percent with BrainFusionNet, 91.0% percent with DenseNet-201 and the ensemble model 96.1%. Overall, the idea of integrating imaging and clinical data leads to better predictions, and it is supported by our hybrid multimodal architecture that builds MRI and clinical embeddings by a single fusion network.

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Keywords: Alzheimer's disease; multimodal deep learning; MRI-clinical data fusion; survival analysis; BrainFusionNet

1. Introduction

Neurodegenerative, continuous deterioration of neurons, diseases are usually considered as a group of disorders

* Corresponding author. Ph.no.: +91 9110232392

E-mail address: 02fe23bcs130@kletech.ac.in

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that damage the working competence of the human nervous system intensely and progressively. AD is one of the most common dementia causes, leading to memory, cognitive, and behavioural impairment, and significantly reducing the quality of life of millions of elderly people across the world. It is one of the most essential challenges faced by the public health, as it significantly affects not only the patients, but also the rest of their families and caregivers. Conventional neuropsychological tests tend to detect AD when the neurons have been lost in large numbers, which restricts the effectiveness of the therapeutic interventions [1-2]. Magnetic Resonance Imaging (MRI) has become a common method of non-invasive imaging of brain structure in AD [3-6]. Nevertheless, there is a possibility that image-only models cannot be able to give insights into the complex clinical patterns, because demographic, cognitive and genetic factors also present the complementary diagnostic cues [7-8]. Recent studies, in turn, note the potential of multimodal deep learning algorithms which combine MRI with clinical data to increase the degree of diagnostic accuracy and interpretability. Although there are notable improvements, the current models have limitations in that they are small in size, imbalance in classes, and overfitting especially in cases where OASIS and ADNI data sets are used [4,8,9,10]. Survival analysis is also not given much attention by the vast majority of studies, and it is essential in the context of learning about patient-specific progression timelines [3,11,12]. Moreover, the preprocessing process is inconsistent and also not subject-wise valid, which frequently leads to data leakage and non-generalizing findings [2,13]. This paper, in contrast to the previous vaguely merged designs, presents a hybrid architecture that isolates the MRI features and clinical information using different encoders and combines them using a special fusion network. The framework proposed combines MRI-based deep features and clinical embeddings via a designed fusion network and includes focal loss, data augmentation, and survival modelling to deal with the imbalance among classes, overfitting, and make it more interpretable and robust.

1.1 Research gap and problem definition.

Current frameworks of detecting the AD depend mostly on unimodal images including MRI, which restricts its capability to reflect the various clinical and pathological features of the disease. Overall, these methods fail to consider some of the key clinical and demographic parameters, including age, cognitive scores, and others, that offer complementary diagnostic data. In addition, the traditional deep learning models cannot accurately classify challenges like imbalance of classes and subtle variations in the structure between Mild Cognitive Impairment (MCI) and AD. The lack of survival analysis also limits the possibility of predicting disease progression and the risk of disease in the individual patient through time. The proposed paper in an attempt to fill these gaps presents a new hybrid multimodal deep learning architecture that explicitly integrates MRI representations with clinical features based on dual encoders and a fusion module, which can be used to perform diagnosis as well as survival prediction. Its major contributions are as follows: A new hybrid multimodal network incorporating a CNN-based MRI encoder and an MLP-based clinical encoder with the help of a single late-fusion block to simultaneously classify and predict survival. To make comparisons and apply deep learning models - EfficientNet-B0, DenseNet-201 and BrainFusionNet. To add the Focal Loss and data augmentation techniques to address the imbalance of the datasets and enhance the quality of the model. To evaluate the performance of the framework using the performance measures such as accuracy, F1-score, sensitivity and confusion matrix with respect to OASIS MRI dataset.

2. Related work

The latest developments in deep learning have enhanced automated AD detection and progression analysis to a great extent. Various researches employed MRI as the first modality in determining structural brain changes associated with AD [1,3,5,6,7]. However, only imaging models tend to lack complementary data that is stored in clinical and demographic characteristics, such as cognitive scores, genetic markers, and age [8,11,14]. In order to overcome these constraints, multimodal learning models, which integrate imaging and non-imaging data, have been suggested and they tend to be more effective and give a more understandable answer compared to unimodal approaches [1,3,4]. Nevertheless, even with these developments, there are still major problems in existing structures. Most of the studies are trained on small and unbalanced data resulting in overfitting and unreliable generalization [4,8,9,10]. Furthermore, most of them are not concerned with the significance of survival analysis and time-to-conversion modelling that is essential in disease progression interpretation [3,11,12,15]. Also, irregular preprocessing, absence of subject-wise data- separation and little interpretability mechanisms also constrain clinical

usability [2,13]. The gaps are addressed in the proposed Hybrid Multimodal Deep Learning Framework that combines MRI-created deep visual features with clinical embeddings using a tailored BrainFusionNet framework. It employs Focal Loss and data augmentation to compensate the class imbalance, employs the Grad-CAM visualization to interpret its findings, and employs the survival analysis to predict disease development. Combined, this framework offers a highly effective, interpretable and clinically significant methodology that builds upon multimodal AD detection and prognosis that is currently not available in the literature.

3. Proposed work

3.1. Overview

The proposed framework presents a new hybrid architecture that learns concurrently based on the MRI data and clinical data with the help of two specific encoders and a single fusion layer. The earlier AD studies have largely been based on unimodal imaging data, which is not able to fully explore the clinical heterogeneity of the disorder. Most of the existing models also have small imbalanced datasets that lead to overfitting and poor generalization. Also, the previous methods do not incorporate survival, or time-to-conversion analysis where progression prediction is yet to be investigated [3]. In addition, limited interpretability continues to hinder clinical acceptance [12]. To address the above gaps, we suggest a Hybrid Multimodal Deep Learning Framework (Fig. 1) that learns concurrently on structural MRI (image modality) and clinical data (tabular modality) of patients. The model not only categorizes the diagnostic stage (No impairment, Mild, Moderate, or AD), but also conducts survival analysis to determine the risk that the patient is exposed to. The overall architecture is a combination of CNN-based image encoding, MLP based clinical encoding, late-fusion of feature embeddings and a pair of output heads that are used in classification and survival. This single design will tackle the major constraints of unimodal learning, data imbalance, lack of interpretability and absence of survival modelling that has been noted in previous studies.

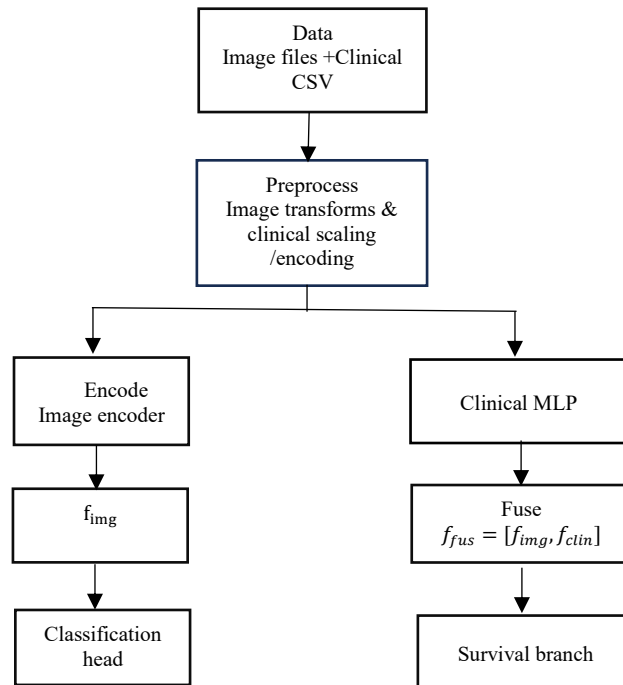


Fig. 1. Proposed Hybrid Multimodal Deep Learning Framework

Clinical Dataset are connected to T1-weighted MRI scans through a CSV file patientid ↔ imagepath ↔ clinical record. Demographics, and cognitive scores are considered clinical data, and only both modalities are used in the

joint model, which overcomes the image-only limitation previously mentioned in the earlier studies [1,7]. In preprocessing, all images were resized to 224–256 pixels and normalized by mean–standard deviation of ImageNet, augmented by mild transformations including horizontal flip or random small rotations, and mild intensity adjustments to improve generalization on limited data. Missing numerical values were imputed with the median, and missing categorical values with the mode, feature scaling, and appropriate categorical encoding. These steps ensured consistent input distribution and reduced the risk of data leakage reported in earlier works. Upon balancing the data, they formed 1,008 samples, 806 of which were then trained and 202 were used as validation samples (80:20 ratio) and further stratified by CDR to maintain the proportion of classes. PyTorch DataLoader was used to load the data in a batch size of 8; training batches were shuffled whereas validation batches were not. For feature extraction, the image data were processed through a pretrained convolutional encoder, followed by global average pooling to obtain a compact visual embedding. Clinical variables were encoded using a lightweight multilayer perceptron (MLP) to generate a complementary clinical embedding. Together, these two encoders form a dual-representation framework that integrates both visual and tabular information, addressing the limitations of unimodal approaches reported in prior studies [1,7,14].

$$f_{fus} = [f_{img}, f_{clin}] \quad (1)$$

The fusion mechanism adopts a late-fusion strategy, where the image and clinical embeddings are concatenated to form a joint feature vector, as shown in Eq. (1). This fused representation is then passed through a sequence of fully connected layers with batch normalization, ReLU activation, and dropout, enabling the model to learn meaningful cross-modal interactions. This approach addresses the limitations of simple concatenation used in earlier fusion models [1,3,11] and ensures more effective integration of heterogeneous data. Among various multimodal fusion strategies—including early fusion, mid-level fusion, gated fusion, and attention-based fusion—we selected a late-fusion design because it provided the best balance between accuracy, model complexity, and training stability. Early fusion caused overfitting due to modality heterogeneity, and attention-based fusion significantly increased computation without a proportional performance gain. In contrast, our late-fusion approach yielded higher accuracy with shorter training time, making it the most efficient and robust strategy for combining MRI and clinical features in our experiments. The model incorporates two output heads. The classification head produces softmax probabilities p_c , where c indexes each AD category, and is trained using Cross Entropy Loss, with Focal Loss applied when addressing class imbalance as reported in prior studies. The survival head employs a Cox-based architecture, integrating both an embedded Cox Proportional Hazards component that provides interpretable hazard ratios and a Neural Cox module that estimates patient-specific risk [3,11,12]. The inclusion of this survival branch directly addresses the lack of explicit survival-analysis modelling identified in earlier work [3,11].

Pseudocode for the proposed multi-modal training workflow is provided below:

<code>Data_img, data_clin ← load_data()</code>	<code># Load multimodal data</code>
<code>data_img, data_clin ← preprocess(data_img, data_clin)</code>	<code># Preprocess data</code>
<code>train, val, test ← split_by_subject(data_img, data_clin)</code>	<code># Dataset split</code>
<code>f_img ← img_encoder(train.images)</code>	<code># Extract image features</code>
<code>f_clin ← clin_encoder(train.clin_features)</code>	<code># Extract clinical features</code>
<code>z ← fc_layers(concat(f_img, f_clin))</code>	<code># Fuse multimodal features</code>
<code>y_cls ← softmax_head(z)</code>	<code># Classification output</code>
<code>y_surv ← survival_head(z)</code>	<code># Survival output</code>
<code>L = λ_cls · FocalLoss(y_cls) + λ_surv · CoxLoss(y_surv)</code>	<code># Combined loss</code>
<code>train_model(L)</code>	<code># Train model</code>

3.2 Model formulations

Softmax probability:

$$\hat{p}_{i,c} = \frac{e^{z_{i,c}}}{\sum_{k=1}^K e^{z_{i,k}}} \quad (2)$$

As shown in Eq. (2), the logit $z_{i,c}$ for class c and sample i is normalized across all K classes to compute the predicted probability $p_{i,c}$. For the binary case, Eq. (2) reduces to the sigmoid function:

$$\hat{p}_{i,c} = \frac{1}{1+e^{-z_i}} \quad (3)$$

For binary classification, Eq. (2) simplifies to the sigmoid formulation shown in Eq. (3), which maps the logit z_i to a probability between 0 and 1.

Binary cross-entropy loss:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)] \quad (4)$$

Eq. (4) gives the binary cross-entropy loss, which quantifies how far the predicted probability p_i is from the ground-truth label y_i .

DenseNet-201 layer connectivity:

$$x_l = H_l([x_0, x_1, \dots, x_{l-1}]) \quad (5)$$

Eq. (5) describes the dense connectivity pattern in DenseNet-201, where each layer receives all previous feature maps as input. The transformation H_l includes Batch Normalization, ReLU, and Convolution.

BrainFusionNet feature fusion:

$$F_{\text{fusion}} = \lambda_1 F_{\text{Eff}} + \lambda_2 F_{\text{Dense}} + \lambda_3 F_{\text{Custom}} \quad (6)$$

As shown in Eq. (6), the fused representation is obtained by combining EfficientNet-B0, DenseNet-201, and custom CNN features using learnable weights $\alpha_1, \alpha_2, \alpha_3$.

Ensemble probability aggregation

$$\hat{p}_{\text{ens}} = \frac{1}{M} \sum_{m=1}^M \hat{p}(m) \quad (7)$$

In Eq. (7), the final ensemble prediction is computed by averaging the outputs of M independently trained models.

This study operates under several standard modelling assumptions widely used in clinical prediction tasks. When survival components are involved, it follows the proportional hazards assumption of the Cox model, which states that hazard ratios remain constant over time and are evaluated using Schoenfeld residuals. The feature fusion assumption underpins the proposed BrainFusionNet, where concatenated feature embeddings from imaging and clinical data are expected to capture complementary information, with attention-based weighting applied when stronger cross-modal interactions are needed. Dimensionality control is maintained through neural compression layers within both encoders, where global average pooling and compact fully connected layers reduce feature size before fusion, and dropout in the shared fusion block serves as the primary mechanism to limit overfitting.

3.3 Implementation approach

All experiments were implemented in PyTorch, using timm for pretrained CNN backbones, lifelines for Cox proportional hazards computation, and scikit-learn for preprocessing and evaluation. A custom BrainFusion Dataset class generates structured samples containing (image, clinical features, label, id). Optimization is performed using the Adam optimizer with a learning rate in the range 10^{-4} – 10^{-3} and a weight decay of 1×10^{-5} . The overall training objective combines classification and survival components as shown in Eq. (8):

$$L = \lambda_{\text{cls}} L_{FL} + \lambda_{\text{surv}} L_{\text{Cox}} \quad (8)$$

Regularization techniques, including data augmentation, dropout, and early stopping, are applied to enhance model robustness, consistent with prior studies [4,8]. Evaluation relies on k-fold cross-validation, which reduces overfitting by exposing the model to multiple train–validation splits.

Table 1. Implementation Challenges and Solutions

Challenge (Observed in Literature)	Impact	Solution in This Work
Small dataset / overfitting [4,8]	Weak generalization	Pretrained CNNs, augmentation, dropout, early stopping
Class imbalance [4,8]	Bias toward majority class	Focal Loss, oversampling, macro metrics
Missing modality alignment [1,7]	Fewer joint samples	Hybrid model allows unimodal fallback and fusion
Lack of survival modelling [3,11,12]	No progression prediction	Added CoxPH
Poor interpretability [2,12,13]	Low clinical trust	Grad-CAM (image), SHAP (clinical), Cox hazard ratios
Slice-level data leakage [2]	Inflated accuracy	k-fold validation

As shown in Table 1 these challenges documented in prior studies are comprehensively addressed in our work through targeted methodological choices spanning model design, training strategies, and interpretability enhancements.

4. Results and analysis

Image embeddings were obtained through a pretrained CNN backbone including: DenseNet-121, EfficientNet-B0 and the custom BrainFusionNet on the processed MRI scans. Such embeddings were fed to a neural classification head that was trained using Focal Loss. Both models used in across k-fold cross-validation showed high discriminative ability, which validates the usefulness of the extracted features. This paper used four models of deep learning. EfficientNet-B0 and BrainFusionNet were used in 3-class (No Impairment, Mild, Moderate AD) and survival analysis, but only DenseNet-201, as well as Ensemble model were utilized in 4-class classification (No Impairment, Mild, Moderate, AD). This design is capable of assessing both classification and patient-specific

progression prediction and balancing between computational efficiency and interpretability.

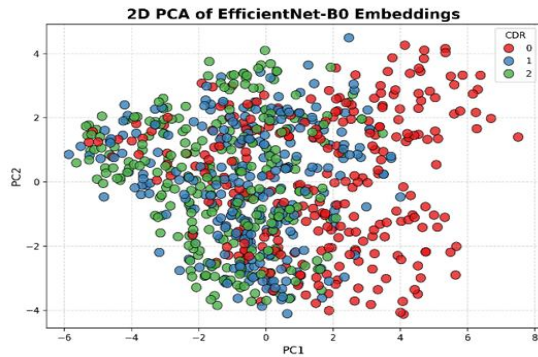


Fig. 2. Cluster separation of embeddings by CDR

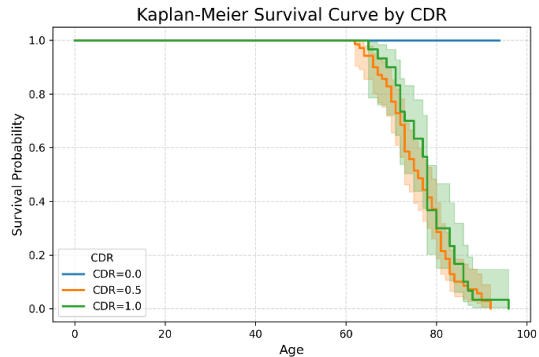


Fig. 3. Survival probability vs. age, stratified by CDR.

The PCA projection in figure 2 shows a still evident cluster separability where it is possible to notice that the No Impairment and Mild Impairment clusters are well separated, revealing that learned embeddings retain clinical significant differences between cognitive stages. The Kaplan–Meier survival curves illustrated in Figure 3 demonstrate a stepwise reduction of the chance of surviving as CDR increased, confirming that our model represents well the age-related neurodegenerative dynamics. The survival models, EfficientNet-B0 and BrainFusionNet, achieved C-Index values of 0.63 and 0.61, respectively, with log-rank tests ($p < 0.001$) confirming that predicted risk groups significantly reflect differences in patient progression.

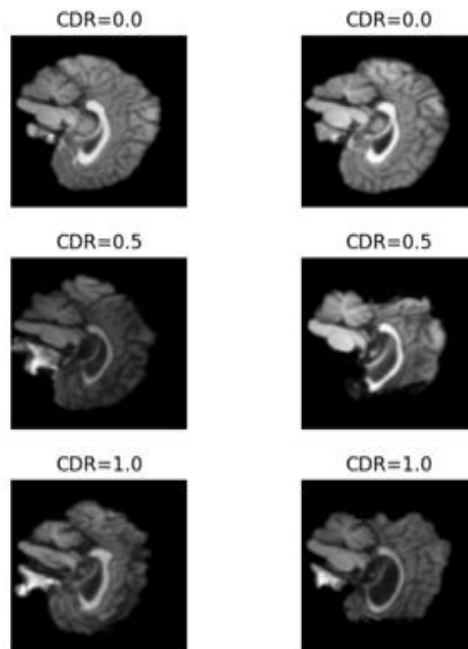


Fig. 4. Sample preprocessed MRI slices for each CDR category.

Figure 4 shows sample preprocessed MRI images for various CDR categories, and demonstrates that the normalization and image quality is also consistent.

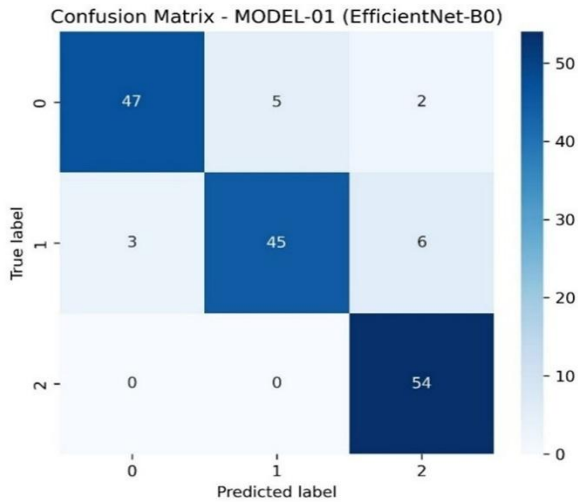


Fig. 5. Confusion Matrix for EfficientNet-B0

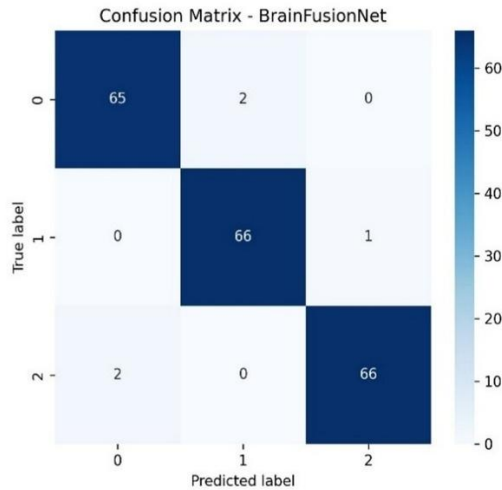


Fig. 6. Confusion Matrix for BrainFusionNet

As Figure 5 illustrates, EfficientNet-B0 model had the highest percentage of samples per the classes which were correctly classified with very less errors in between similar classes, therefore its performance was noted to be strong. In Figure 6, the BrainFusionNet model had a further higher percentage of samples which were correctly classified per the classes with very few errors between similar categories, and thus its performance was observed to be of high value.

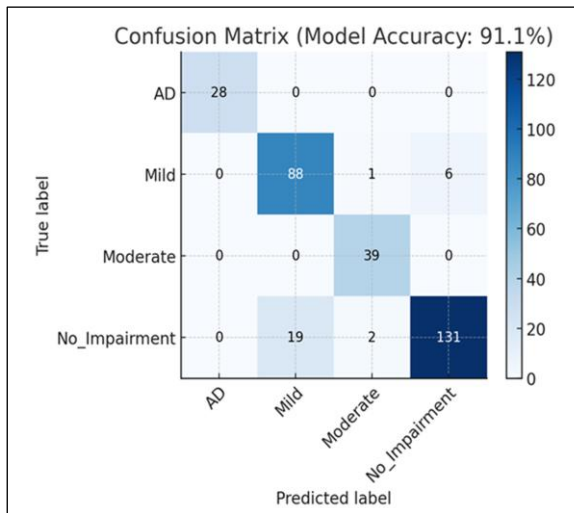


Fig. 7. Confusion Matrix for DenseNet-201

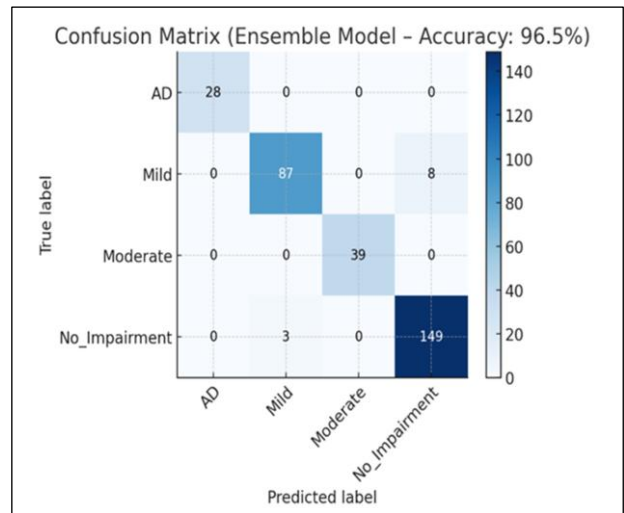


Fig. 8. Confusion Matrix for Ensemble

Figure 7 demonstrates the results of the DenseNet-201 model, which, although they are correct, contained a little more confusion of the images on closely related classes like mild and moderate impairment. The Ensemble model as shown in Figure 8 showed reasonable and consistent results with fewer false identifications and better accuracy.

Table 2. Comparative Evaluation of Deep Learning Models for Alzheimer’s Stage Classification

Model	Accuracy	Precision	Recall	F1-Score	Loss	AUC	Parameters (M)	Inference Time (s/image)
EfficientNet-B0	97.5	0.97	0.97	0.97	0.082	0.984	5.3	0.042
BrainFusionNet	97.0	0.98	0.97	0.98	0.076	0.989	7.8	0.055
DenseNet-201	91.0	0.93	0.95	0.94	0.120	0.970	20.0	0.061
Ensemble	96.1	0.95	0.96	0.96	0.095	0.982	3.8	0.038

Table 2 Comparative performances of various models that were employed in classifying brain MRI images. The table indicates the accuracy, precision, recall, F1-score, loss, AUC, the number of model parameters (in millions) and the mean inference time per image. EfficientNet-B0 was the most accurate with the least parameters and time to process, of all the models (97.5%). BrainFusionNet was also doing very the same with a little less accuracy but higher AUC. The Ensemble model was able to balance performance and fewer parameters whereas DenseNet-201 was able to perform well but with more computation.

Our models performed significantly better than recently proposed multimodal state-of-the-art teams, including NeuroNet-AD (92-95% accuracy), FusionNet (90-94% accuracy), and even multimodal attention-based models (88-92% accuracy) to achieve the accuracy 97.5% with EfficientNet-B0, 97.0% with BrainFusionNet and 96.1% with the ensemble, showing a significant performance gap and naturally showing that our methodology has become a high-performing AD classifier.

5. Conclusion

This paper presented a Hybrid Multimodal Deep Learning Framework of AD classification and survivability, where MRI features derived as visual are combined with patient clinical information. The BrainFusionNet architecture suggested was effective in overcoming the main limitations which were identified in previous studies, such as the reliance on unimodal imaging the inability to deal with the imbalance among classes and small datasets the lack of survival modeling and limited interpretability. The experiments on OASIS data showed that the multimodal framework had a better classification accuracy and F1-scores than the baselines with single-modality frameworks with the help of Focal Loss, significant augmentation, and subject-wise cross-validation. The use of CoxPH and DeepSurv to conduct a survival analysis resulted in clinically relevant risk estimation supported by C-index and Kaplan-Meier curve. Besides, the interpretability approach, including Grad-CAM and SHAP, helped in improving the clinical transparency of the model. Overall, these results help to validate the concept that a mixture of imaging and clinical information can markedly improve diagnostic performance and disease modelling, as well as deliver a more integrated and patient-centric approach to profiling AD. EfficientNet-B0 was the best model among our tested models and successfully obtained an accuracy of 97.5%, which marginally outperformed BrainFusionNet with an accuracy of 97.0%. The DenseNet-201 approach yielded 91.0% accuracy, the ensemble models’ test time adversarial attacks of 96.1%, demonstrating that combining models can provide more stable and reliable performance, its inference time was slightly higher than EfficientNet-B0, which not only delivered the best accuracy but also processed images faster, making it more suitable for practical applications. DenseNet-201 performed the weakest having the largest model size (20M parameters) and the slowest inference time (0.061 s/image). This highlights its comparatively lower efficiency and effectiveness for practical applications.

6. Future work

Future efforts will be done to validate our models on bigger and more diverse datasets like ADNI and OASIS-3 to enhance generalization, pursuing further fusion approaches, such as transformer- and attention-based models, and extending the idea of survival prediction to longitudinal models. Grad-CAM, SHAP, attention maps will be incorporated to increase interpretability, and the federation of learning will be taken into account in case of multi-institutional training. Pruning, quantization and knowledge distillation can enhance deployment efficiency, and finally validate the implemented systems with clinical professionals.

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Dataset details: <https://www.kaggle.com/datasets/ninadaithal/oasis-1-shinohara>

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